Université Libre de Bruxelles

Faculté des Sciences

Département de Physique



Proton structure

Generating the transverse distributions from first principles

Link to the up-to-date version : https://github.com/Arthur-42/Memoir/blob/main/out_dir/memoire.pdf

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Résumé

Abstract

Introduction

Idées principales, contexte, intérêt, déroulé

Citations temporaires pour que tout apparaisse dans la bibliographie : [7], [4], [6], [2], [1]

Chapter 1

[Elements of QFT]

Quelques mots sur les concepts de base de la QFT (champs, particules, méthode perturbative, règles de Feynman, algèbre, modèle standard). Variation d'α_s (explications sur la renormalisation en annexe). Confinement.

Chapter 2

Proton structure and the parton model

In this chapter, we will introduce the central element of this thesis : the internal structure of the proton, how to probe it and how to express it mathematically, within the bounds of perturbative QCD. We will begin with a very brief history of the study of the proton through scattering, to put emphasis on the role of the energy scale on the information accessed. This will bring us up to deep inelastic scattering (DIS), who provided evidence of the inner constituent of the proton, then termed "partons".
Finally, we will see that a deeper look of the parton model with the tools of perturbative QCD lead to the study of the parton distribution functions (PDF) and their energy scale dependence through the Dokshitzer-Gribov-Lipatov-Altarelli-Parisi (DGLAP) equation. The goal of this chapter is really to introduce the PDF and the DGLAP equation, as they are essential for all that will follow.

The following topics are extensively covered in most textbooks on perturbative QCD. The references used here are [2], [3], [6], [1].

2.1 Early results from electron-proton scattering

2.2 Deep inelastic scattering

Chapter 3

A simplified parton evolution

In this chapter, we will show how to numerically compute PDFs and obtain transverse momentum distributions with a parton branching method, with a very simplified framework, taken from the exercises adjoining [4]. Even if this implementation is far less sophisticated than the one this thesis is focused on, it already holds many of its key points, and will allow us, by fiddling with its parameters, to investigate its behaviour and reveal points of interest to keep in mind for the full implementation.

Note : *This is the reason why completing the exercises from [4] was the first step of my assignment. The parton evolution presented here is a slight variation of exercise 3 of these lectures. I used [4] and [6] as a guide to complete it.*

3.1 Implementation

We want to evolve a starting gluon density $f(x, \mu_0^2) = 3(1-x)^5/x$ from scale $\mu_0^2 = 1\,\text{GeV}^2$ to scale $\mu_f^2 = 10\,000\,\text{GeV}^2$. We will also generate their transverse momentum k_t during the evolution, and assuming a starting distribution with zero mean and $\sigma = 0.7\,\text{GeV}$. We will use the splitting function

$$P_{gg}(z) = 6\left(\frac{1}{z} + \frac{1}{1-z}\right) \tag{3.1}$$

with cut-offs $z_{min} = \epsilon$, $z_{max} = 1 - \epsilon$ with $\epsilon = 0.1$. As a last simplification, we will take the strong coupling to be constant : $\alpha_s = 0.1$.
We will also redo the same exercise, with a quark-quark splitting

$$P_{qq}(z) = \frac{4}{3}\frac{1+z^2}{1-z} \tag{3.2}$$

and assuming the same starting distribution (which could pass as an approximation of a sea quark density, but not a valence quark density).

In order to evaluate the density at the final scale, $f(x, \mu_f^2)$, we will generate 1 000 000 events from the starting density and evolve them by simulating their possible branching according to the known probability densities. The evolved events will then be distributed following $f(x, \mu_f^2)$. During the evolution of each event, we will also generate the k_t of the emitted gluon, and compute the change of k_t of the emitter, to obtain a k_t distribution at the final scale.

For each event, we must perform the following steps :

1. **Generate the starting momentum fraction x_0 and transverse momentum $k_{t,0}$:**

Here, we generate x_0 with the importance sampling method, using the approximate function $g(x) = \frac{1}{x \log(x_{max}/x_{min})}$, with $x_{min} = 0.000\,01$ and $x_{max} = 1$.
Then, drawing a uniform random number $R_{x_0} \in [0, 1]$, we get x_0 and its weight w_{x_0} as

$$\boxed{x_0 = x_{min} \left(\frac{x_{max}}{x_{min}}\right)^{R_{x_0}}} \tag{3.3}$$

$$\boxed{w_{x_0} = 3\,(1 - x_0)^5 \, \log(x_{min}/x_{max})} \tag{3.4}$$

We generate $k_{t,0}$ using the central limit theorem, by summing 12 uniform random numbers $R_i \in [0, 1]$, with then

$$\boxed{k_{t,0} = 0.7 \left(\sum_{i=1}^{12} R_i - 6\right)} \tag{3.5}$$

These will indeed distribute following a Gaussian distribution of mean 0 and standard deviation 0.7. We also need the two components $k_{x,0}$ and $k_{y,0}$ for later use, so we draw a uniformly distributed azimuthal angle $\phi \in [0, 2\pi]$ to initialise them at $k_{x,0} = k_{t,0} \cos(\phi)$, $k_{y,0} = k_{t,0} \sin(\phi)$.

2. **Generate the next branching scale μ_i^2**

We generate the next scale μ_i^2 with respect to the Sudakov form factor

$$\Delta_S(\mu_{i-1}^2, \mu_i^2) = \exp\left(-\int_{\mu_{i-1}^2}^{\mu_i^2} \frac{d\mu^2}{\mu^2} \int_{z_{min}}^{z_{max}} dz \frac{\alpha_s}{2\pi} P_{gg}(z)\right) \tag{3.6}$$

by drawing a uniformly distributed random number $R_{\mu_i^2} \in [0, 1]$ and inverting the relation $R_{\mu_i^2} = \Delta_S(\mu_{i-1}^2, \mu_i^2)$. In our case, it is easy since the only scale dependence is in the $1/\mu^2$ term, as we are approximating α_s to be constant. Thus we have

$$\begin{aligned}
&\ln(R_{\mu_i^2}) = -\int_{\mu_{i-1}^2}^{\mu_i^2} \frac{d\mu^2}{\mu^2} \int_{z_{min}}^{z_{max}} dz \frac{\alpha_s}{2\pi} 6 \left(\frac{1}{z} + \frac{1}{1-z}\right) \\
\Leftrightarrow \quad &\ln(R_{\mu_i^2}) = -\frac{\alpha_s}{2\pi} \ln\left(\frac{\mu_i^2}{\mu_{i-1}^2}\right) 6 \left[\ln\left(\frac{z_{max}}{1 - z_{max}}\right) - \ln\left(\frac{z_{min}}{1 - z_{min}}\right)\right] \\
\Leftrightarrow \quad &\ln\left(\frac{\mu_i^2}{\mu_{i-1}^2}\right) = -\frac{\pi}{3\alpha_s} \ln\left(\frac{z_{max}(1 - z_{min})}{(1 - z_{max}) z_{min}}\right)^{-1} \ln(R_{\mu_i^2}) \\
\Leftrightarrow \quad &\boxed{\mu_i^2 = \mu_{i-1}^2 \, R_{\mu_i^2}^{-\frac{\pi}{3\alpha_s} \ln\left(\frac{z_{max}(1-z_{min})}{(1-z_{max})z_{min}}\right)^{-1}}}
\end{aligned} \tag{3.7}$$

With the quark-quark splitting function $P_{qq}(z)$ from Equation 3.2, the exponent becomes

$$-\frac{3\pi}{2\alpha_s} \left(\frac{z_{min}^2 - z_{max}^2}{2} + (z_{min} - z_{max}) + 2 \ln\left(\frac{1 - z_{min}}{1 - z_{max}}\right)\right)^{-1} \tag{3.8}$$

We then compare the new scale with the final scale : if $\mu_i^2 > \mu_f^2$, this means that no branching occurred up to the final scale, and we keep the last x_i and $k_{t,i}$ generated. If not, then a branching occurred at this scale, and we carry on with the following steps to generate its effect.

3. **Generate the splitting variable z_i and the new $k_{t,i}$**

The splitting variable is generated in a similar way, using the normalised integrated splitting function as the cumulative function. So, with a uniformly generated random number $R_{z,i} \in [0,1]$, we have

$$\begin{aligned}
R_{z,i} &= \frac{\int_{z_{min}}^{z_i} dz\, P_{gg}(z)}{\int_{z_{min}}^{z_{max}} dz\, P_{gg}(z)} \\
\Leftrightarrow \quad R_{z,i} &= \frac{6\left(\ln\left(\frac{z_i}{1-z_i}\right) - \ln\left(\frac{z_{min}}{1-z_{min}}\right)\right)}{6\left(\ln\left(\frac{z_{max}}{1-z_{max}}\right) - \ln\left(\frac{z_{min}}{1-z_{min}}\right)\right)} \\
\text{so with} \quad & \boxed{A = R_{z,i} \ln\left(\frac{z_{max}}{1-z_{max}}\right) + (1-R_{z,i}) \ln\left(\frac{z_{min}}{1-z_{min}}\right)} \\
\Leftrightarrow \quad \frac{z_i}{1-z_i} &= e^A \\
\Leftrightarrow \quad & \boxed{z_i = \frac{e^A}{1+e^A}}
\end{aligned} \tag{3.9}$$

For the quark-quark splitting of Equation 3.2, inverting the relation is more tricky. We prefer to use importance sampling again, using as an approximate function $g(z) = \frac{1}{(1-z)\ln(1-z_{min}/1-z_{max})}$ with thus

$$z_i = 1 - (1-z_{min})\left(\frac{1-z_{max}}{1-z_{min}}\right)^{R_{z_i}} \tag{3.10}$$

with weight

$$w_{z_i} = \frac{4}{3}(1+z_i^2)\ln(1-z_{min}/1-z_{max}) \tag{3.11}$$

We can then update the momentum fraction by multiplying the previous one with the new splitting variable :

$$x_i = z_i\, x_{i-1} \tag{3.12}$$

In the quark case, we also need to multiply the weight : $w_{x_i} = w_{z_i}\, w_{x_{i-1}}$ while in the gluon case, the weight stays at its initial value from Equation 3.4.

We can also generate the new $k_{t,i}$. Actually, we generate the k_t of the emitted particle, then calculate the resulting $k_{t,i}$ of the emitter. The choice of expression for the k_t is not set in stone, we have to make a heuristic choice. Here, we will investigate two options : *transverse ordering* and *angular ordering*. The former prescribe

$$k_t = \mu_i \tag{3.13}$$

Where the natural scale for the k_t is then taken to be the current energy scale. The latter uses

$$k_t = \mu_i (1 - z_i) \tag{3.14}$$

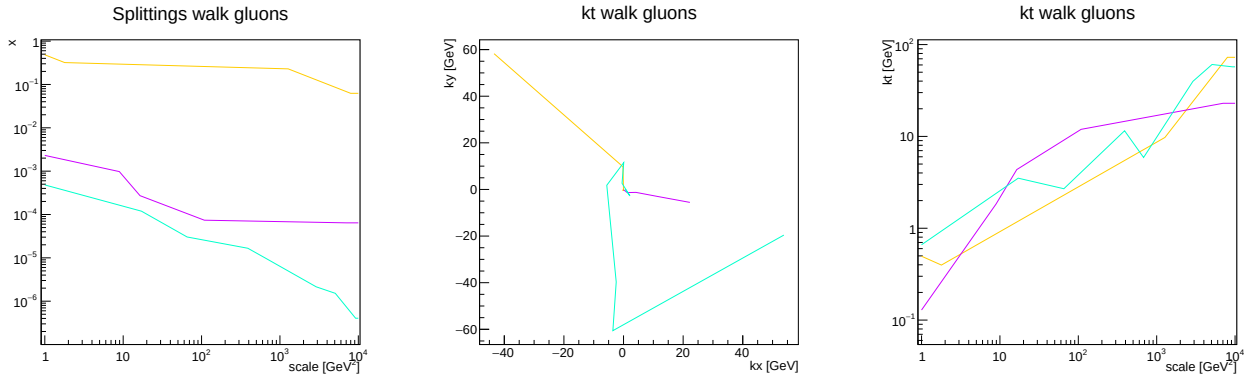
Which rather associate the energy scale with the angle of the momentum of the emitted particle, with respect to the one of the emitter. The discussion on how these are obtained is deferred to the next chapter. In the meantime, we will just use them as-is to investigate the resulting distributions.
In the following analysis, we will try angular ordering first, and transverse ordering second.

Generating a uniformly distributed azimuthal angle $\phi \in [0, 2\pi]$, we can finally evaluate

$$\boxed{k_{x_i} = k_{x_{i-1}} - k_t \cos\phi \quad ; \quad k_{y_i} = k_{y_{i-1}} - k_t \sin\phi \quad ; \quad k_{t_i} = \sqrt{k_{x_i}^2 + k_{y_i}^2}} \tag{3.15}$$

4. **Jump back to 2. until the end of the loop**

Going through these steps looks like this (with angular ordering for the k_t generation) :



(a) sample gluon evolution (b) sample gluon k_t generation (c) sample gluon k_t evolution

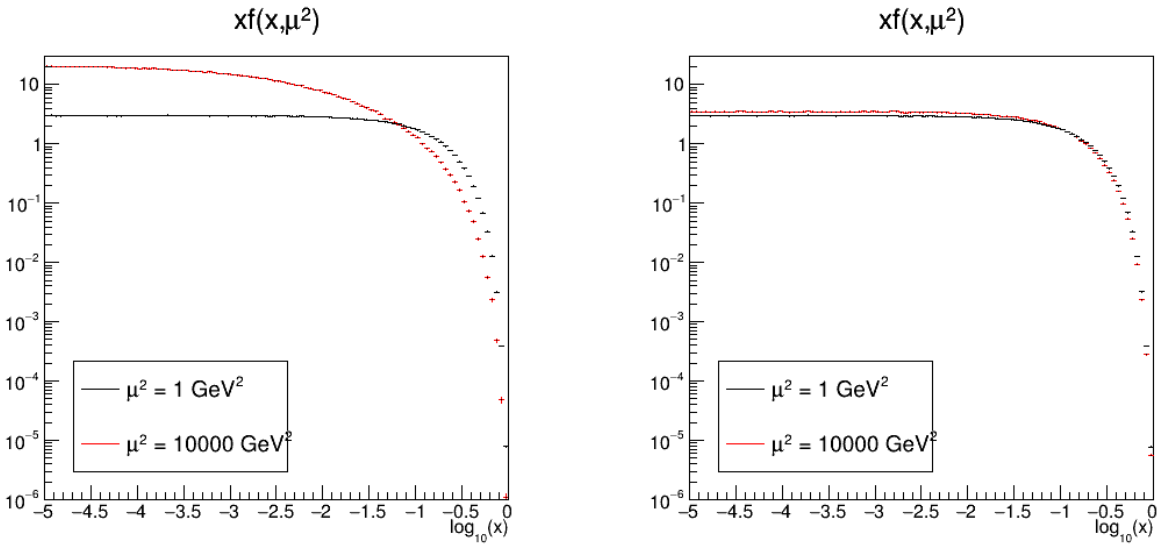
Figure 3.1: Evolution process for a sample of three gluon.

On Fig. 3.12a, we can see the momentum fraction being reduced as splittings occur through the scale evolution. Fig. 3.12b shows the evolution of the x and y components of the transverse momentum as branchings occur, and Fig. 3.1c plots the k_t with respect to the scale.

To get a significant sample, we perform this for 1 000 000 events, and plots the resulting densities in histograms. Note that in order to obtain histograms for the weighted density $x f(x, \mu^2)$, we need to multiply the weight for each x by x itself. There is one last thing to take into account to get the correct normalisation : this method evolves as many partons as that are generated for the initial distribution, but their momentum fraction only lowers. This seems in conflict with the fact that the integral of the momentum density must of course be conserved. It is so because the momentum fraction is not lost, but spread between more partons. To account for this, we must simply normalise the evolved densities to the same momentum integral as the initial ones. We can also note that only the $f(x, \mu^2)$ integrals are conserved through evolution, not the $x f(x, \mu^2)$ nor the k_t ones. (The former because of the shift in the distribution of x, the latter because of the increase of the number of partons : more high-k_t partons does not automatically indicate less low-k_t ones).

3.2 Results

Here are the resulting histograms :



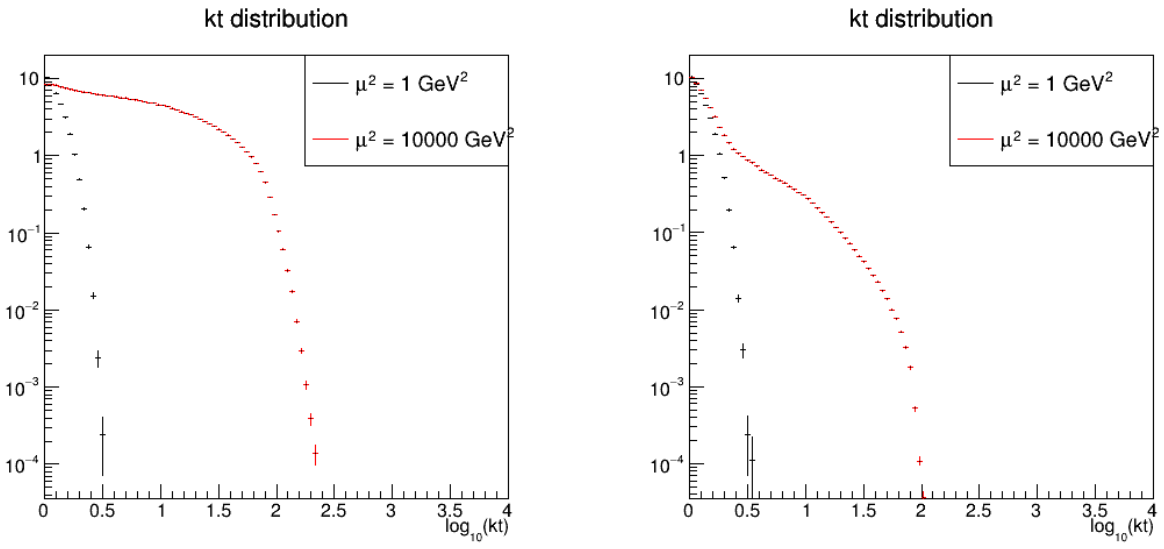
(a) gluon density

(b) quark density

Figure 3.2: Evolution of the momentum weighted parton density.

In the gluon case, the migration to small momentum fractions x is plain to see, while in the quark case it is much less pronounced. Seeing the latter, we might fear an inadequacy in the parameters used, perhaps in the cut-off. We will see that the cut-off does play a role, but is also counterbalanced. This result is actually robust, in a sense that we will explore a bit later. The difference between the results for the gluon and the quark densities really come from the behaviour of their splitting functions.

Here are the results for the transverse momentum k_t (with angular ordering) :



(a) gluon distribution

(b) quark distribution

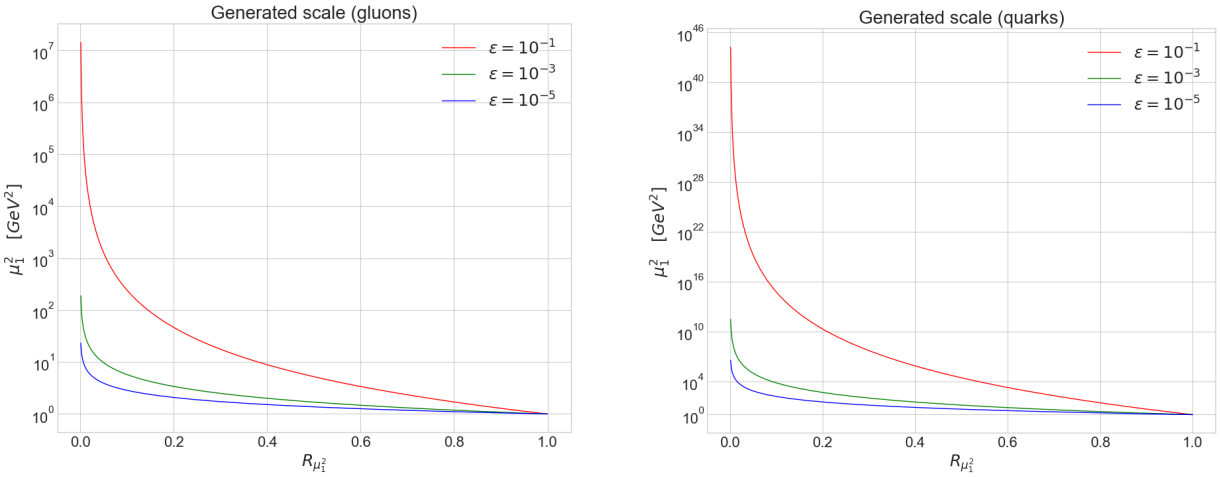
Figure 3.3: Evolution of the transverse momentum distribution.

We can notice that the distributions decrease sharply because of the scale limitation, although it does exceed it in the gluon case, likely because of the accumulated steps. The quark distribution being notably lower might put in conjunction with the less pronounced evolution, but the abrupt stop at the final scale might indicate that

hardly more than one branching ever occur for a single event.

This justifies an investigation on the dependence of these results on, say, the cut-off ϵ, which gives $z_{min} = \epsilon$, $z_{max} = 1 - \epsilon$ and thus control the integrals on the splitting function.

Let us take a look at the ϵ dependence for the scale generation :

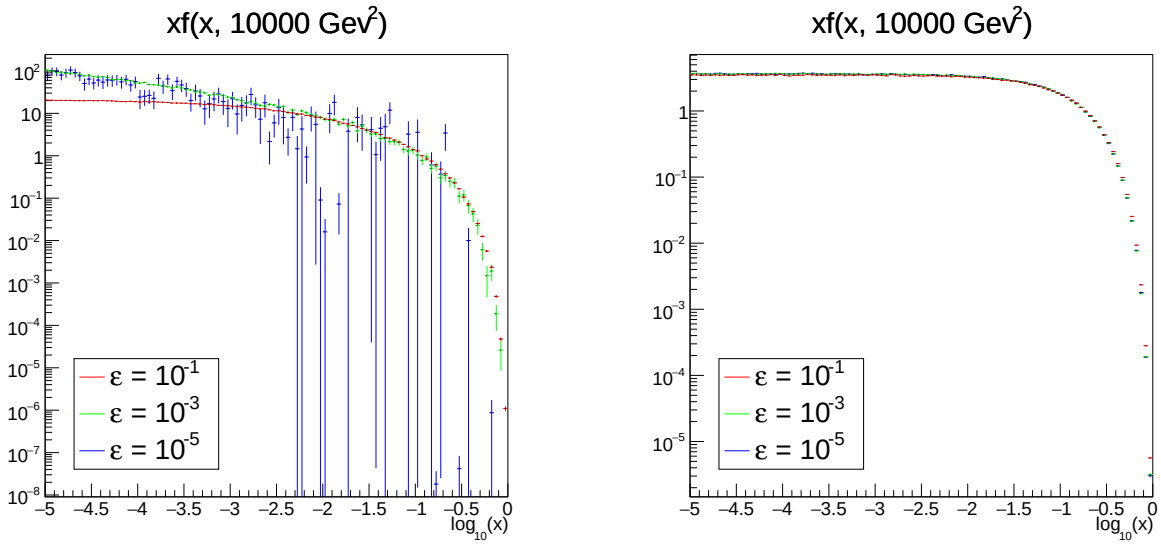


(a) gluon case (b) quark case

Figure 3.4: First generated scale μ_1^2 with respect to the random number $R_{\mu_1^2}$, depending on ϵ.

Let us recall that the evolution is stopped when $\mu_i^2 > \mu_f^2 \equiv 10^4\,\text{GeV}^2$. Let us also recall that from Equation 3.7, with a different initial scale $\mu_0^2 \neq 1$ we would have got the same graph, with μ_1^2 multiplied by μ_0^2, so this graph is representative of the generation of scales after the first.
We can see that decreasing ϵ decrease the probability of generating large scales (a smaller range of $R_{\mu_1^2}$ would produce them), which overall means that there will be more branching before the final scale is reached. We can also note that the scales generated in the quark case are often very large, especially for $\epsilon = 0.1$. In this latter case, for more than half the events, no branching would occur at all (as for $R_{\mu_1^2} \in (0, 0.54)$, we have $\mu_1^2 > \mu_f^2$).

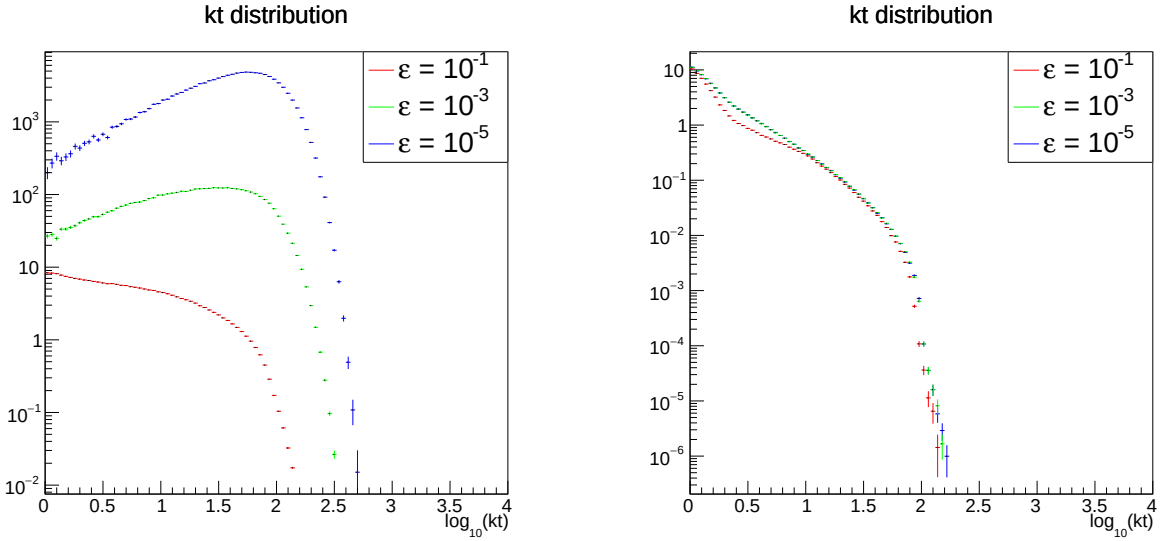
Let us have a look at how ϵ affects the density plots :



(a) gluon density (b) quark density

Figure 3.5: Evolved momentum weighted parton density, depending on ϵ.

The gluon density is strongly affected : decreasing ϵ strongly increase the density for small momentum fractions.
Note : *The $\epsilon = 10^{-5}$ case is messy because of scarcely filled bins : on 1 000 000 events generated, only 76 end up in the right half of the plot (the majority is actually in the underflow).*
Perhaps surprisingly, the quark density seems mostly unaffected by the variation of ϵ. We need to look at the k_t distribution to find a difference :

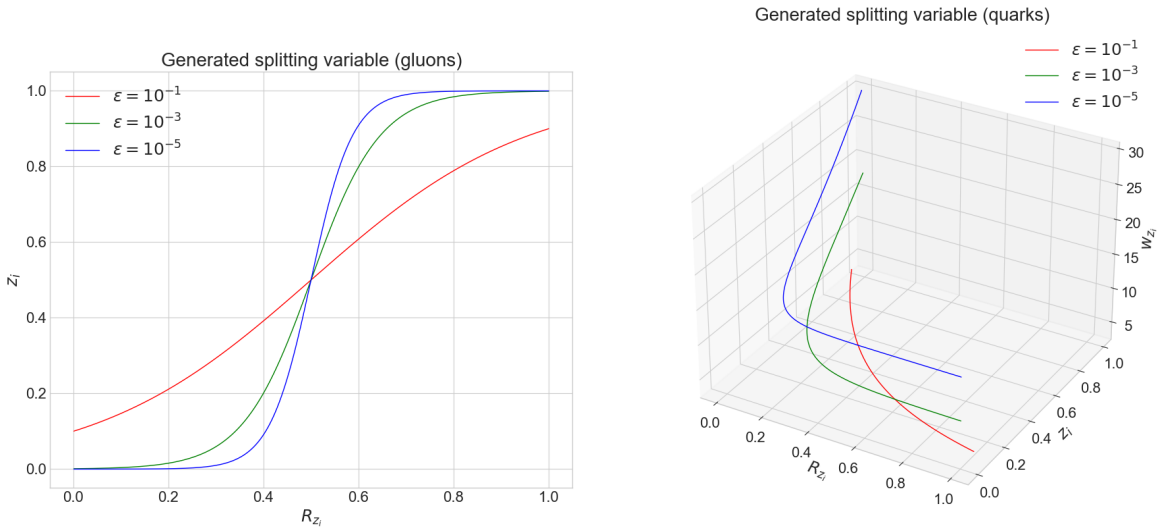


(a) gluon distribution

(b) quark distribution

Figure 3.6: Evolved transverse momentum distribution, depending on ϵ.

The gluon density is still strongly affected, while for the quark density, the $\epsilon = 10^{-3}$ and $\epsilon = 10^{-5}$ straighten up a bit from the $\epsilon = 10^{-1}$ case, but match with each other. We can suppose that with ϵ to large, there weren't quite enough branchings generated to get a statistically sound k_t distribution, while the quark distribution otherwise don't seem to depend on it. This is not the case for the gluon distributions. To understand this better, we need to take a look at the ϵ dependence of the splitting variable generation :



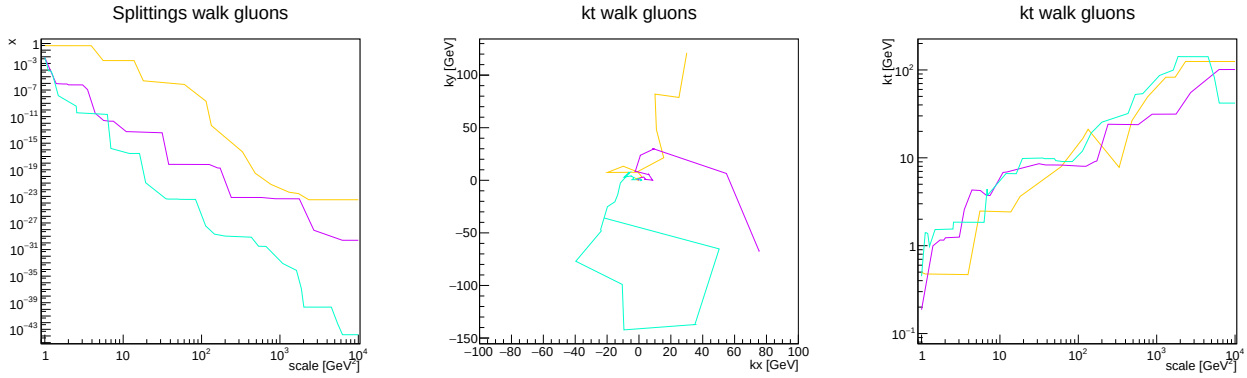
(a) gluon case

(b) quark case, showing the weights

Figure 3.7: Generated splitting variable z_i with respect to the random number R_{z_i}, depending on ϵ.

Since we used the importance sampling method to generate the quark splitting, we need to show the corresponding weights to accurately display the importance of each value in the results. We can notice that for the quark case, reducing ϵ increase the importance of large z_i values. Thus events where the splittings doesn't

change the momentum fraction much (z close to 1) weight more in the histogram, while, as seen in Fig. 3.4, more branchings occur and produce more splittings, reducing x more. Apparently, these two opposite effects cancel each other in the quark case, giving ϵ-independant results (provided ϵ is not *too* large). This is quite fortunate, since the cut-off is unphysical.
We are not so lucky with the gluon density, but we can see why in Fig. 3.7a : decreasing ϵ makes z_i close to 1 more probable, but also z_i close to 0. This latter effect explain the strong migration towards very small momentum fractions in Fig. 3.5a. We also generate more splittings, and given the formula for the k_t used in Equation 3.14, we make more large jumps in k_t, which explains the large density for large k_t in Fig. 3.13a.
Compared to Fig. 3.12, we now get more events like these :

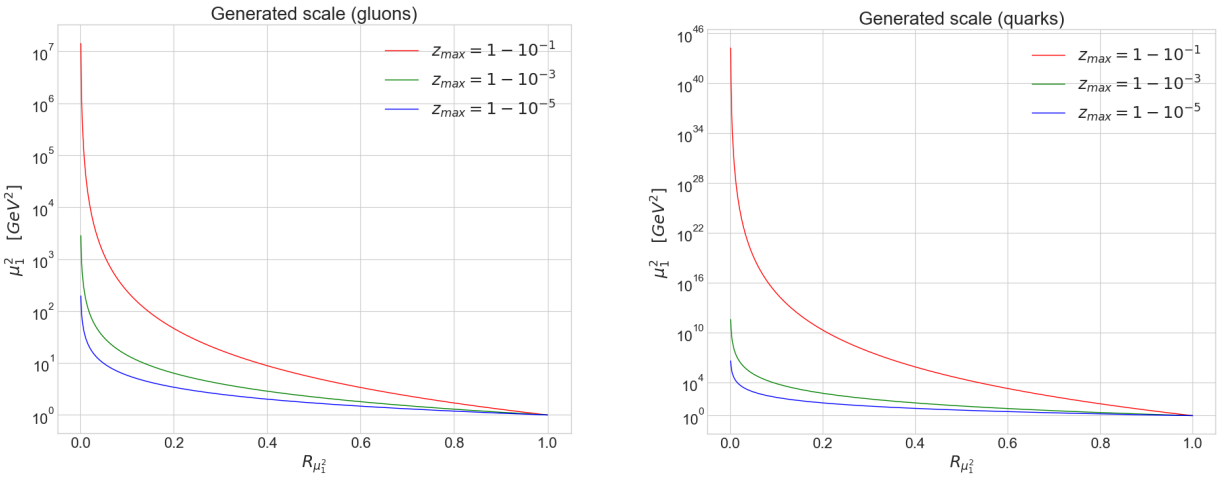


(a) $\epsilon = 10^{-5}$ gluon evolution (b) $\epsilon = 10^{-5}$ gluon k_t generation (c) $\epsilon = 10^{-5}$ gluon k_t evolution

Figure 3.8: Evolution process for a sample of three gluon, with $\epsilon = 10^{-5}$.

These events exemplify well the behaviour discussed : we can see on Fig. 3.8a a lot of splitting who sharply decrease the momentum fraction, and on the remaining plots, the tendency to build a large k_t.

We can then ask ourselves how would this turn out if we fixed z_{min}, in hope to avoid this small z_i issue. With $z_{min} = 0.1$ fixed, the variable generation would then behave this way :

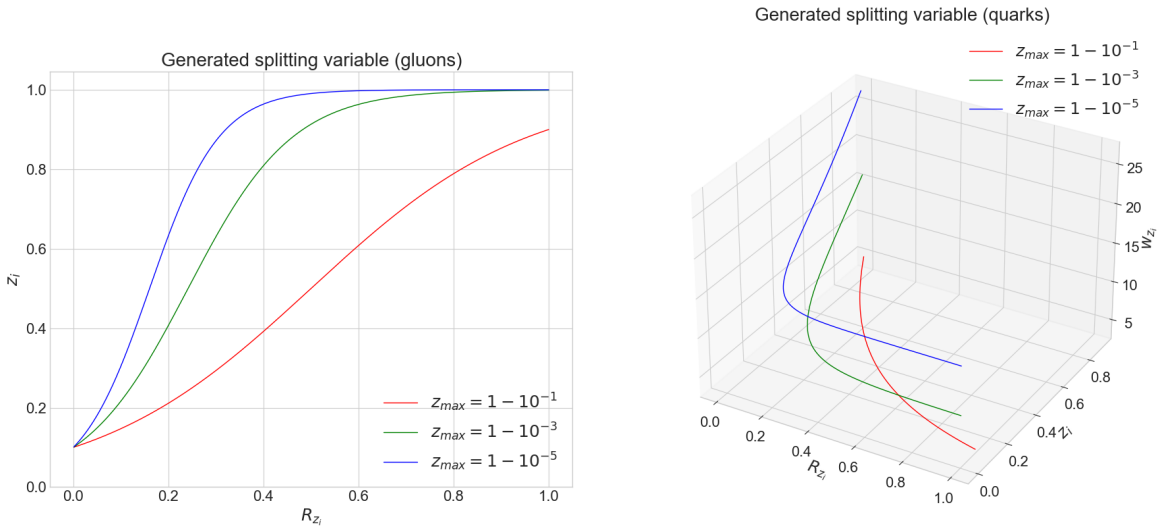


(a) gluon case (b) quark case

Figure 3.9: First generated scale μ_1^2 with respect to $R_{\mu_1^2}$, fixed z_{min}, variable z_{max}.

The generation of the scale is noticeably changed only for the gluon case. This is natural, as the splitting function for the gluon in Equation 3.1 diverge as $z \to 0$, while it is not the case for the quark in Equation 3.2. By fixing z_{min}, we avoid this divergent contribution. So now we have more chances to generate bigger scales, compared to the unfixed z_{min} scales, so we shouldn't get as many splittings per events as we could see in Fig. 3.8a. (To be clear, approaching z_{max} of 1 still increases the amount of splittings, just not as much as when

we approached z_{min} from 0 too.)
The change for gluons is especially striking for the splitting variable generation :

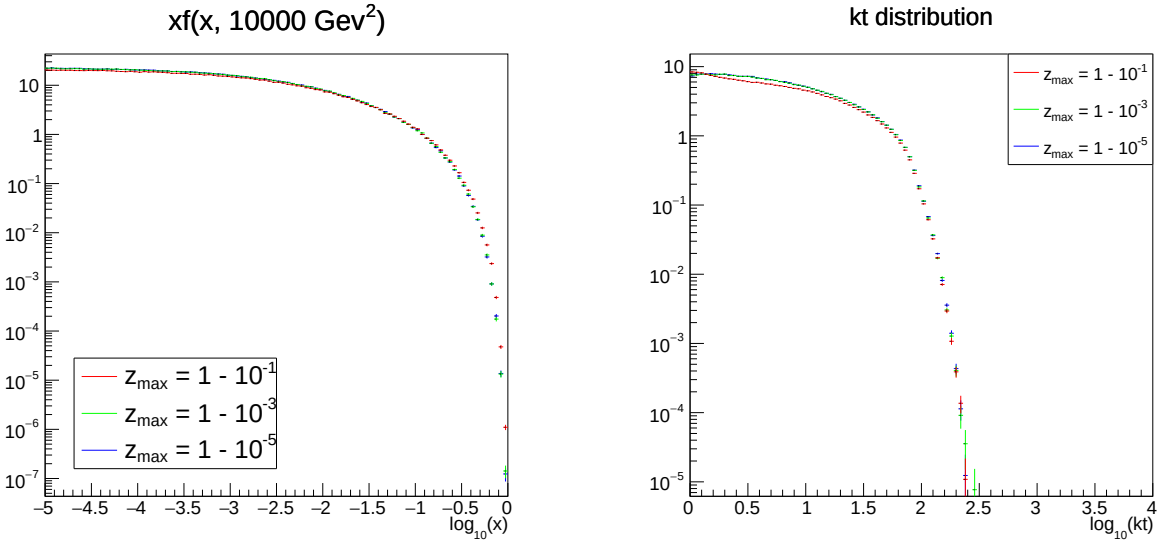


(a) gluon case

(b) quark case, showing the weights

Figure 3.10: Generated splitting variable z_i with respect to R_{z_i}, fixed z_{min}, variable z_{max}.

Now, we can't generate z values arbitrarily close to zero, and small values are becoming less likely as we take z_{max} closer to 1. The splittings will then be less impactful.
For the quarks, the biggest change is in the weight, but it is proportional for each case. Fixing $z_{min} = 0.1$ doesn't change much the results for the quarks, while it very much do for the gluons :



(a) momentum weighted density

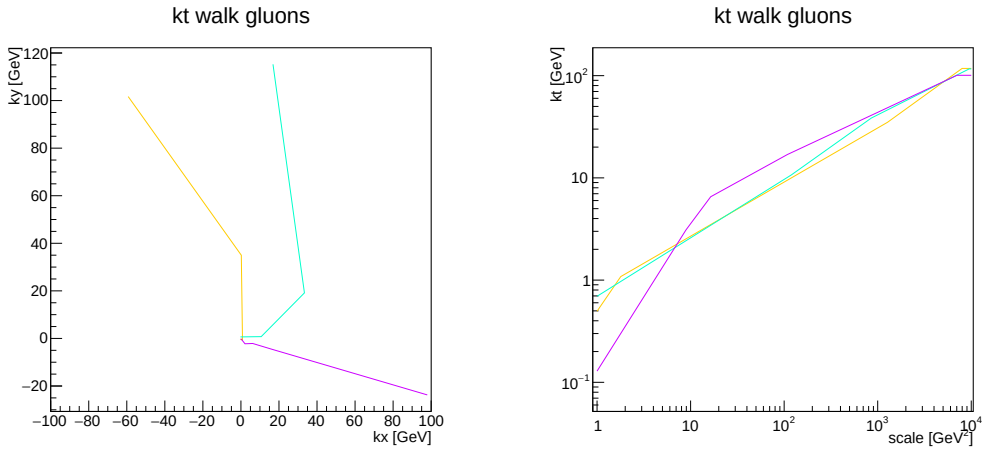
(b) transverse momentum distribution

Figure 3.11: Evolved gluon densities, fixed z_{min}, variable z_{max}.

We finally get stable results. We saw here that we can hope to get consistent results independently of the arbitrary choice of z_{max} (provided that it is reasonably close to 1). This is because bringing z_{max} closer to 1 increase the amount of splittings generated, but also increase the probability to generate the splitting variable z_i closer to 1, thus making the splittings less impactful, for both the x and the k_t distribution.

Now we can investigate what happens when we choose transverse ordering (see Equation 3.13) for the k_t generation. It doesn't affect the x density, but it should of course change the k_t densities. Firstly, here is an

example of gluon event generation, with $z_{min} = 0.1$ and $z_{max} = 0.9$:

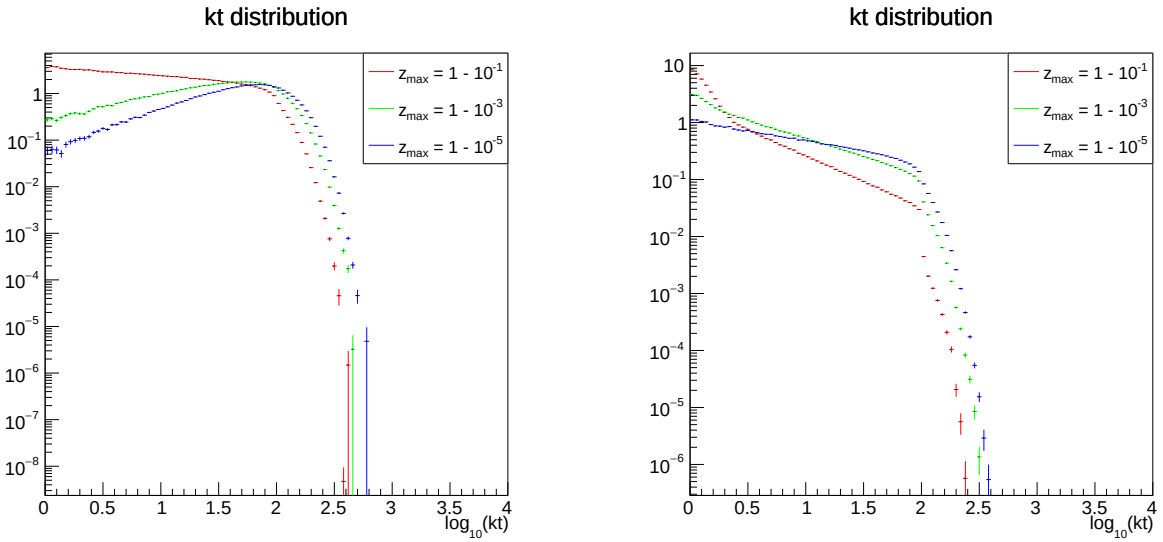


(a) sample gluon k_t generation

(b) sample gluon k_t evolution

Figure 3.12: k_t evolution with transverse ordering, for a sample of three gluons, $z_{max} = 0.9$.

Now successive steps always do a larger k_t jump. But we can already anticipate an issue here. The amount of splitting is sensitive to z_{max}, and so is the z_i generation, but the k_t generation is only sensitive to the former, since Equation 3.13 does not depend on z_i. We can expect to lose z_{max} independence, and this is indeed the case :



(a) gluon distribution

(b) quark distribution

Figure 3.13: Evolved k_t distribution, transverse ordering, variable z_{max}

Angular ordering is a better choice, because it correctly compensates its sensitivity to the amount of splittings with its z_i dependence. We will see in the next chapter another physical argument that could have helped guess its suitability.

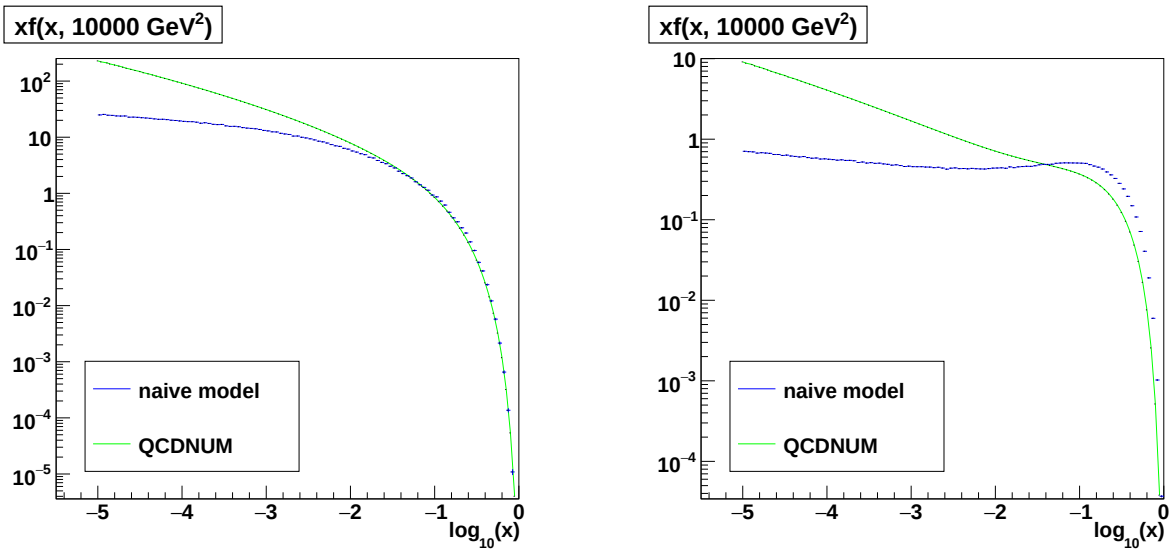
Of course, the model used here is too simple to yield physically accurate results. It is nonetheless interesting to see how far (or how close) we can get. Here is a comparison of our toy model with PDFs produced with QCDNUM with NNLO computations, both starting with the following realistic parton distributions

$$\text{gluons :} \quad x f(x, \mu_0^2) = 1.7\, x^{-0.1}(1-x)^5 \tag{3.16}$$

$$\text{down (valence)}: \quad x f(x, \mu_0^2) = 3.064320\, x^{0.8}(1-x)^4 \tag{3.17}$$

$$\text{down (sea)}: \quad x f(x, \mu_0^2) = 0.1939875\, x^{-0.1}(1-x)^6 \tag{3.18}$$

evolved from $\mu_0^2 = 2\,\text{GeV}^2$ to $\mu_f^2 = 10^4\,\text{GeV}^2$, and where for our "naive" model we took $z_{min} = 0.1$, $z_{max} = 1-10^{-5}$:



(a) gluon density

(b) down quark (valence + sea) density

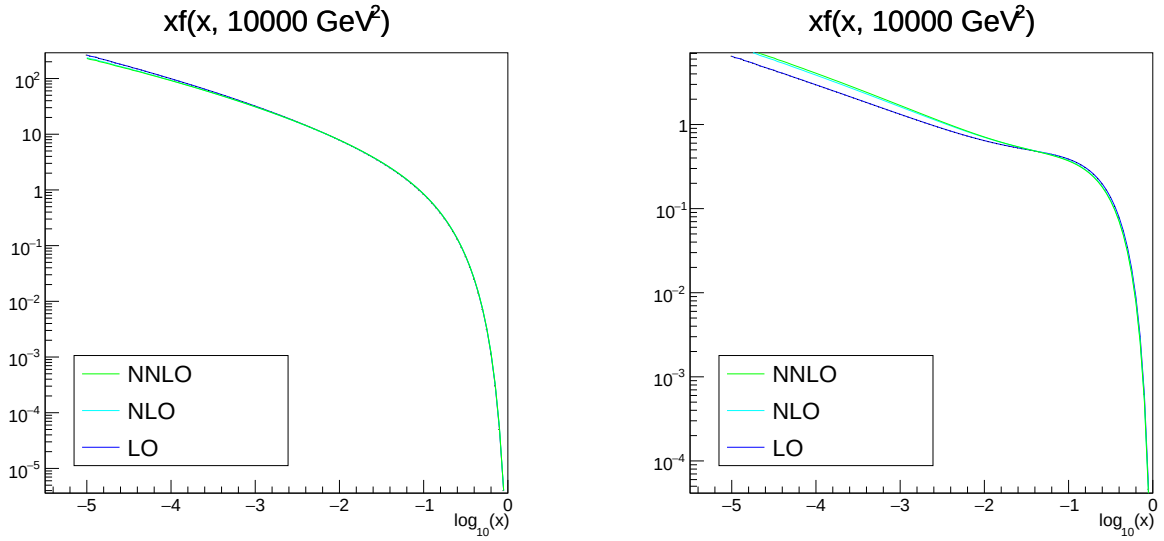
Figure 3.14: Comparison of momentum weighted parton densities.

We can see that our naive model does not evolve the distributions as far as QCDNUM does. This is not a surprise.

First, we took $z_{min} = 0.1$, which is quite high, and so strongly reduces the overall impact of the splittings. We had to restrict ourselves to this here, since we saw at Fig. 3.13a that lowering z_{min} lead to bad k_t distributions. A more accurate model should allow us to use a considerably lower z_{min}.

Second, we only do one kind of splitting (gluon-gluon or down-down). But several other forms of splittings are allowed, notably gluon-quark, next-to-leading-order splittings with loops, splittings with other quarks flavours, and so on. Restricting us to one single kind of splitting of course limits the magnitude of the evolution.

And finally, we only took leading order splitting functions, and we did not evolve the coupling α_s, which would also affect the splitting process. To get a sense of how much the order of computations would change the results, we can see how it affects those of QCDNUM :



(a) gluon density

(b) down quark (valence + sea) density

Figure 3.15: Comparison of momentum weighted parton densities of QCDNUM.

On Fig. 3.14 we see the evolved densities of QCDNUM, where the splitting function and the running of α_s are calculated at different orders. There is no straightforward way to make these computations with QCDNUM without running α_s, so we cannot test the effect of this here. Looking back at Equation 3.7, we can see that, since a running α_s would have higher values at low scale that the one used in our naive model, it would have a higher probability to generate low scales, and thus induce more splittings. (Even without looking at Equation 3.7, we can guess that a higher coupling makes interactions – here splittings – more probable).
Clearly the order of the splitting functions is not the dominant reason for the difference between the QCDNUM results and the ones of our naive model. However, we expect that higher-order computations for the parton branching method would allow us to use a more realistic (i.e. smaller) z_min and still give a good-looking k_t distributions (unlike at Fig. 3.13a, as stated above), and that would make a large step towards good results.

We can note on Fig. 3.14a that our naive model is still not so bad for the gluon density at high x values. This is probably because the splittings involving gluons are dominant, and so gluon-gluon splitting only would be valid at high x, while omitting quark-gluons splittings in Fig. 3.14b would be too much of an overlook.

3.3 Summary and perspectives

Résumer les étapes, expliquer ce qui changera dans l'application complète. Rôles différents de z_{min} et z_{max}, trouver une bonne justification pour la valeur de z_{min}.

Chapter 4

[Study of the DGLAP evolution equation]

Sum rules, reformulation, etc...

Chapter 5

À déterminer+

Fonctionnement du code...

Conclusion

Annexes

A Computation of $P_{qq}(z)$

Here, we compute the LO quark-quark splitting function.
Note : *This is the only explicit computation of a matrix element in this thesis. I deemed it useful to include as it show how to get from the QFT basics one of the splitting functions, which are extensively used in this thesis. It may also serve as a warm-up or dusting-off of the QFT computations skills of the reader (as it surely has been for the author). This idea was taken from [6], where the same is done for $P_{qg}(z)$.*

The corresponding Feynman diagram is :

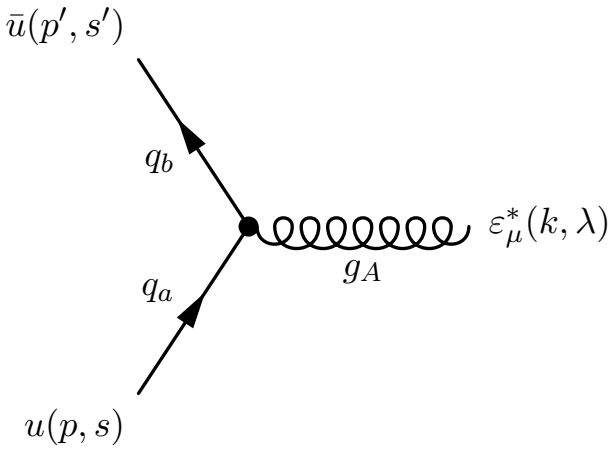
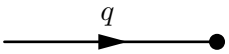
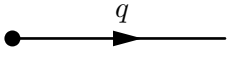
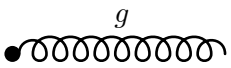
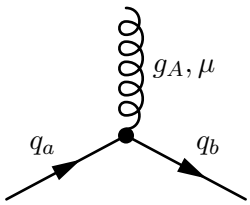


Figure A.1: quark-quark splitting.

The quark momenta are taken to flow in the direction of the fermion arrows, while the momentum of the gluon is taken to be outgoing, so $p'^\mu = p^\mu - k^\mu$.
For the Feynman rules, we follow the convention of [5]. As a matter of fact, the following computation shadows a similar one in Section 17.5 of [5].

	$u(p, s)$	Spinor of an incoming quark of momentum p and spin s.
	$\bar{u}(p, s)$	Spinor of an outgoing quark of momentum p and spin s.

	$\varepsilon^*_\mu(k,\lambda)$	Polarisation quadrivector of an outgoing gluon of momentum k and polarisation index λ.
	$-igT^A_{ba}\gamma^\mu$	Vertex for quarks of colour a and b, and gluon of colour compound A, whose polarisation quadrivector carries μ indices. g is the coupling constant, T^A is a Gell-Mann matrix, and the γ^μ are the gamma-matrices.

The momenta are taken as follows (in the massless quark limit) :

$$p = (p,\, 0,\, 0,\, p) \quad ; \quad p' \approx (zp,\, p_\perp,\, 0,\, zp) \quad ; \quad k \approx ((1-z)p,\, -p_\perp,\, 0,\, (1-z)p) \tag{A.1}$$

That way, we have $p^2 = 0$, $p'^2 = 0 + \mathcal{O}(p_\perp^2)$, $k^2 = 0 + \mathcal{O}(p_\perp^2)$, where the outgoing quark and gluon are thus slightly off-shell. ***Note :*** *the expansion of the adimensionned quantity is actually in powers of* $(p_\perp/p)$*, so we only need* $p_\perp << p$.
If we want to better approximate the case of a off-shell outgoing quark and on-shell outgoing gluon, we may take

$$p' = \left(zp,\, p_\perp,\, 0,\, zp - \frac{p_\perp^2}{2(1-z)p}\right) \quad ; \quad k = \left((1-z)p,\, -p_\perp,\, 0,\, (1-z)p + \frac{p_\perp^2}{2(1-z)p}\right) \tag{A.2}$$

With then $p'^2 = 0 + \mathcal{O}(p_\perp^4)$, $k^2 = -p_\perp^2/(1-z) + \mathcal{O}(p_\perp^4)$. The case of a off-shell gluon and on-shell quark is obtained the same way by substituting $p_\perp^2/2(1-z)p \to p_\perp^2/2zp$ above. Nevertheless, we will stick to the definitions at A.1 for the computation of the matrix element as it yields the same result up to $\mathcal{O}(p_\perp^2)$ terms (thus $\mathcal{O}(p_\perp^3)$ terms for the squared matrix element).

A fermion spinor $u(p,s)$ is represented by a 4-component column vector (and let us remind that the adjoint is $\bar{u} = u^\dagger \gamma^0$). It can be further subdivided in two helicity components, each represented by a 2-component vector, with then $u(p,s) = (u_L(p,s),\, u_R(p,s))^T$. The detailed expression of the left-handed component is

$$u_L(p,s) = \sqrt{p\cdot\sigma}\,\xi^s \quad \xrightarrow[\text{high energy}]{} \quad \sqrt{2p^0}\,\frac{1}{2}(I - \hat{\boldsymbol{p}}\cdot\boldsymbol{\sigma})\,\xi^s \tag{A.3}$$

With the normalised 2-components spinors $\xi^\uparrow = (1,\, 0)^T$ for spin up and $\xi^\downarrow = (0,\, 1)^T$ for spin down. Here $\hat{\boldsymbol{p}}$ is the 3-component spatial part of the momentum vector divided by p^0, and we have introduced the notation $\sigma \equiv (1,\, \boldsymbol{\sigma}) \equiv (1,\, \sigma^1,\, \sigma^2,\, \sigma^3)$ for the Pauli matrices (and $\bar{\sigma} \equiv (1,\, -\boldsymbol{\sigma})$). In helicity basis, the Pauli and gamma matrices are :

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad ; \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad ; \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad ; \quad \gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \tag{A.4}$$

In the massless limit for the quarks, the spin and helicity are linked, so a left-handed quark necessarily has spin down ($u_L(p,\uparrow) = 0$), and a right-handed one has spin up ($u_R(p,\downarrow) = 0$). Then we will do the following computation per helicity cases, and we won't have to sum over the spins s, s'. We can also note that the helicity of the quark must be conserved. For left-handed quarks, we have :

$$u_L(p,\downarrow) = \sqrt{2p}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad ; \quad \bar{u}_L(p',\downarrow) = \sqrt{2zp}\begin{pmatrix} -p_\perp/2zp \\ 1 \end{pmatrix} \tag{A.5}$$

In the massless limit for gluon (which is correct for real gluons, and close enough for the slightly off-shell gluon treated here), only the transverse polarisations are nonzero. We can take the ones with a defined helicity :

$$\varepsilon^*(k,L) = \frac{1}{\sqrt{2}}\left(0,\,1,\,i,\,\frac{p_\perp}{(1-z)p}\right) \quad ; \quad \varepsilon^*(k,R) = \frac{1}{\sqrt{2}}\left(0,\,1,\,-i,\,\frac{p_\perp}{(1-z)p}\right) \tag{A.6}$$

(One can check that these are indeed transverse since $\varepsilon \cdot k = 0$). With all that, we can begin our calculations. The matrix element would be

$$i\mathcal{M} = (-ig)\,\bar{u}_L(p',\downarrow)\,T^A_{ba}\,\gamma^\mu\,u_L(p,\downarrow)\,\varepsilon^*_\mu(k,\lambda) \tag{A.7}$$

Separating with respect to the helicities (let us recall that the helicity is conserved for the quark) :

$$\begin{aligned} i\mathcal{M}(q_L \to q_L\,g_L) &= (-ig)\,u_L^\dagger(p',\downarrow)\,T^A_{ba}\,(-\sigma^i)\,u_L(p,\downarrow)\,\varepsilon_i^*(k,L) \\ &= ig\sqrt{2zp}\begin{pmatrix}\frac{-p_\perp}{2zp}, & 1\end{pmatrix} T^A_{ba}\,\frac{1}{\sqrt{2}}\left[\begin{pmatrix}0&1\\1&0\end{pmatrix}+\begin{pmatrix}0&1\\-1&0\end{pmatrix}+\frac{p_\perp}{(1-z)p}\begin{pmatrix}1&0\\0&-1\end{pmatrix}\right]\sqrt{2p}\begin{pmatrix}0\\1\end{pmatrix} \\ &= ig\,T^A_{ba}\,p\sqrt{2z}\left(-\frac{p_\perp}{zp}-\frac{p_\perp}{(1-z)p}\right) \\ &= -ig\,T^A_{ba}\,\frac{\sqrt{2z}}{z(1-z)}\,p_\perp \end{aligned} \tag{A.8}$$

$$\begin{aligned} i\mathcal{M}(q_L \to q_L\,g_R) &= (-ig)\,u_L^\dagger(p',\downarrow)\,T^A_{ba}\,(-\sigma^i)\,u_L(p,\downarrow)\,\varepsilon_i^*(k,L) \\ &= ig\sqrt{2zp}\begin{pmatrix}\frac{-p_\perp}{2zp}, & 1\end{pmatrix} T^A_{ba}\,\frac{1}{\sqrt{2}}\left[\begin{pmatrix}0&1\\1&0\end{pmatrix}+\begin{pmatrix}0&-1\\1&0\end{pmatrix}+\frac{p_\perp}{(1-z)p}\begin{pmatrix}1&0\\0&-1\end{pmatrix}\right]\sqrt{2p}\begin{pmatrix}0\\1\end{pmatrix} \\ &= ig\,T^A_{ba}\,p\sqrt{2z}\left(-\frac{p_\perp}{(1-z)p}\right) \\ &= -ig\,T^A_{ba}\,\frac{\sqrt{2z}}{(1-z)}\,p_\perp \end{aligned} \tag{A.9}$$

This interaction is invariant under parity change, so the matrix elements are the same under $L \leftrightarrow R$, and thus we already know the results for right-handed quarks.
We can move on to the calculation of the squared matrix element. We sum on the outgoing colours and helicities, and average on the incoming ones. The averaging yields a 1/6 prefactor (one over 2 helicities times 3 quark colours). We have

$$\begin{aligned} &\frac{1}{6}\sum_{A,a,b}\sum_{h,h'}|\mathcal{M}(q_h \to q_h\,g_{h'})\mathcal{M}^\dagger(q_h \to q_h\,g_{h'})| \\ &\quad= \frac{1}{6}\sum_{A,a,b}\left(T^A_{ba}\,T^A_{ab}\right)\left(2\,g^2\frac{2z}{z^2(1-z)^2}p_\perp^2 + 2\,g^2\frac{2z}{(1-z)^2}p_\perp^2\right) \\ &\quad= \frac{4}{3}\frac{2g^2p_\perp^2}{z(1-z)}\left[\frac{1+z^2}{1-z}\right] \end{aligned} \tag{A.10}$$

Where we used that the Gell-Mann matrices are hermitian and that

$$\frac{1}{3}\sum_{A,a,b} T^A_{ba}\,T^A_{ab} = \frac{4}{3} \tag{A.11}$$

Now, we are interested in the "factored" cross section σ_{fact} of this process, in the sense that the outgoing quark is

supposed to connect to another process (say with cross section $\sigma(q'X \to Y)$), but that the gluon branching can be factored from the whole interaction, with cross section $\sigma(qX \to gY) = \sigma_{\text{fact}}\,\sigma(q'X \to Y)$. The validity of this is discussed in [auto-référence à ajouter]. In the expression we seek here, we will have the propagator (squared) of the outgoing quark and its momentum unintegrated, as it is really an internal propagator between the gluon splitting and the process described by $\sigma(q'X \to Y)$. For the expression of the propagator to be nondivergent, we take the outgoing quark to be slightly off-shell, with momentum given as in A.2.
We have :

$$\sigma_{\text{fact}} = 2p'^0\,\frac{1}{2p^0}\int\frac{d^3k}{(2\pi)^3}\frac{1}{2k^0}\left[\frac{1}{6}\sum|\mathcal{M}\mathcal{M}^\dagger|\right]\left(\frac{1}{p'^2}\right)^2 \tag{A.12}$$

The prefactor $2p'^0$ accounts for the fact that the prefactor from the incoming particles in $\sigma(q'X \to Y)$ is $1/(2p'^0 2E_X)$ while the correct one for $\sigma(qX \to gY)$ is $1/(2p2E_X)$, so in σ_{fact} we eliminate the unwanted $1/2p'^0$. The remaining terms above describe the cross section of the gluon splitting, with the outgoing quark momentum unintegrated and the inclusion of its propagator squared, as announced.
Now let us express $d^3k = d|p(1-z)|d^2p_\perp = p\,dz\,\pi\,dp_\perp^2$, use the definition $g^2 = 4\pi\alpha_s$, and use the explicit expressions for the momenta :

$$\begin{aligned}
\sigma_{\text{fact}} &= 2zp\,\frac{1}{2p}\int\frac{p\,dz\,\pi\,dp_\perp^2}{8\pi^3}\frac{1}{2(1-z)p}\left[\frac{4}{3}\frac{8\pi\alpha_s p_\perp^2}{z(1-z)}\left(\frac{1+z^2}{1-z}\right)\right]\frac{(1-z)^2}{p_\perp^4}\\
&= \int_0^1 dz\int_{\mu_0^2}^{\mu_f^2}\frac{dp_\perp^2}{p_\perp^2}\frac{\alpha_s}{2\pi}\left[\frac{4}{3}\left(\frac{1+z^2}{1-z}\right)\right]\\
&= \int_0^1 dz\,\frac{\alpha_s}{2\pi}\,\ln\left(\frac{\mu_f^2}{\mu_0^2}\right)\left[\frac{4}{3}\left(\frac{1+z^2}{1-z}\right)\right]
\end{aligned} \tag{A.13}$$

The transverse momentum integral has been performed from a lower cut-off μ_0^2 (as there physically is : the quark mass, that we had negliged all throughout) to the desired factorisation scale μ_f^2. We have thus found the LO quark-quark splitting function

$$P_{qq}(z) = \frac{4}{3}\frac{1+z^2}{1-z} \tag{A.14}$$

It is the one used for the simple parton evolution method of Chapter 3, notably.
But there is more to say about it. [discussion fonction de branchement compète]

IA use

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